

Yarn Construction

Basic Principles

Yarns are composed of textile fibres. There are different ways in which fibres can be joined together to provide a variety of structures that in turn, create a wide variety of fabrics. A yarn is continuous strand of textile fibre, filament or material in a form suitable for knitting, weaving or otherwise intertwining to form a textile fabric.

Yarn occurs in the following forms;

- a) A number of fibres twisted together
- b) A number of filaments laid together without twist
- c) A number of filaments laid together with more or less twist
- d) A single filament - monofilament
- e) One or more strips made by a lengthwise division of a sheet of material such as a natural or synthetic polymer, or paper or a metal foil, used with or without twist in a textile construction.
- f) The varieties possible include single, plied, cord, thread, cabled and fancy yarns.

Spun and Filament Yarns

- A spun yarn is composed of short staple fibres that are twisted together, resulting in a fuzzy yarn with protruding fibre ends.
- Spun yarns of cotton are most common in apparel. Also common are yarns of polyester cotton blend.
- A filament yarn is composed of long fibres that are grouped together or strictly twisted together.
- If filament yarns are used the yarns may be either multifilament, composed of several filaments or monofilament composed of a single filament.

Yarn Processing

- Yarns composed of both staple and filament length fibres may be identified by various terms depending on the process used in their manufacture.
- Traditional systems used to convert staple fibres into yarns, whether natural or man-made, are cotton systems and woollen or worsted system.
- These are grouped together under one general heading ring spinning. Other non-conventional methods used today include open end spinning or break spinning, etc.

Filament Yarns

- The manufacture of yarns from filament fibre (natural or man-made) is a simple and direct process. Silk is the only natural filament fibre, and the size of silk depends on the number of cocoons reeled at one time.
- Filament yarns of man-made are made either continuous or discontinuous process. In the continuous process the number of filaments in the final yarn and the number of openings in the spinneret are the same. The filaments are extruded, drawn to develop strength molecular orientation and fineness, and then combined with the amount of twist desired and wound around take-up bobbins.
- The discontinuous process differs in that the filament, without twist and for some types without drawing are wrapped onto packages or cones.
- When needed these filaments are wound from the package then twisted and drawn if necessary.
- The yarn is then wrapped onto bobbin beams ready for use in making fabrics.
- Filaments yarns are smooth unless they are deliberately made regular for novelty effects, such as bulk-yarn or textured-bulk filament yarns.

Staple – Fibre Yarns

Ring Spinning; Cotton System

The following chart summarizes the steps in producing a spun yarn.

The Cotton System

Operation	Purpose
Opening	Looses, blends, cleans, forms lap.
Carding	Cleans, straightness, forms carded sliver.
Drawing	Parallels, blends, forms drawn sliver.
Combing	Parallels, removes short fibres forms combed sliver. (used for long staple cotton only)
Roving	Reduces, sizes, inserts twists, and forms a roving.
Spinning	Reduces size twists, winds the finished yarn on bobbins.
Winding	Rewinds yarn from bobbins to spools or cones

Opening Cleaning and Blending

- Staple fibres arrive at the yarn processing plant in large bales.
- To make yarns fibres must be similar length and relatively uniform so that the spun yarn can be of uniform quality.
- To accomplish this fibres from a variety of production lots, fields or animals must be blended together.
- The same fibre from each bale is fed into the opener and blender.
- It is important to separate or open the fibre mass to a single fibre state or as close to that as possible. The opening and blending machines separate the fibres and blend fibres from different bales.
- The technique used differs according to the type of opening or blending machine used.
- The opening operation separates the fibre into a loose fluffy mass.
These loose fibres are fed into a hopper where a measure amount is laid on a conveyor belt and delivered to the picking unit where additional blending occurs.
- The picker further opens cleans and blends fibres through a system of rollers and forced air.
- The blend of fibres is blown onto a collecting cylinder to form a layer
 $\frac{1}{2}$ " - $\frac{3}{4}$ " (1.3-1.9cm) in thickness.
- As the cylinder rotates the layer of fibre is rolled off to form the picker lap for carding.
- During this operation most of the dirt and impurities that might have been present are removed by either gravity or centrifugal force.
- Cotton fibre receive more opening and blending than man-made fibres
Since they have more impurities and greater variations than do man-made fibres.
- No matter which system used the quality of the final yarn is dependent
Largely on the selection of fibres and on the thoroughness of opening, cleaning, blending, and picking operation.

Carding

- Carding continues the cleaning of the fibres. It removes fibres too short for use in yarns. The process partially aligns the fibres so that their longitudinal axis is somewhat parallel.
- Carding is accomplished by wire cards or granular cards.
- Wire cards contain two layers of card clothing consisting of wire flats (rectangular shapes) in which fine wire pins are anchored.

- The flats are attached to a steel cylinder and to an endless belt that rotates over the top portion of the cylinder.
- This thin web is gathered into a soft mass and pulled into a rope like strand of fibres called a sliver.
- This sliver is pulled through a con-shaped outlet and doffed or delivered to cans or to a conveyor belt.
- The card sliver is not completely uniform in diameter and the fibres are somewhat random in arrangement.

Drawing

- Drawing increases the parallelism of the fibres and combines several carded slivers into one drawn sliver.
- This is blending operation that contributes to greater yarn uniformity
- Drawing is done by sets of rollers each set running successfullly faster than the preceding set.

Combing

- If long staple fibres are to be spun carding and drawing are followed by combining.
- The fundamental purpose of combining is to parallel fibres and to remove any short fibres from the long staple so that the combed fibres are more uniform in length.
- During the combining operation, as much as 20% the fibre may be removed.
- Fibres emerge from the combed sliver
- Combing is done to achieve a yarn that is superior to a carded yarn in smoothness, fineness, evenness and strength.
- For cotton or cotton blends, the term combed yarn is used.
- For wool or wool blends the term worsted yarn is used.

Roving

- Reduces the drawn sliver, increases parallelism of fibres and inserts a small amount of twist to give strength.
- The product is called a roving.
- The drawn sliver is fed between sets of rollers
- Each set rotates faster than the set behind it, the front about ten times faster than the back set of rollers. This pulls the fibres out, reduces diameter of strand and paralyses the fibres.
- The roving is wound onto a bobbin.

- Full bobbins are removed from the frame and delivered to the spinning frame.

Spinning

- The final process in ring spinning of simple yarns in the spinning operation.
- During spinning the roving is alternated to the desired diameter called the final draft and the desired amount of twist inserted.
- The roving is fed down into the spinning area where it feeds between sets of roller
- Like in roving the front set of rollers rotates faster than the back set.
- The alternated yarn is fed down guided through a U-shaped guided called a traveller, which moves around the take-up package bobbin on a ring, hence the name "ring spinning".
- The spindle turns about 13000 revolutions ten minutes; the traveller is slightly lower.
- About 11 meters of yarn is wound onto the bobbin per minute.

Ring Spinning; Woollen and Worsted Systems

Woollen yarns are carded and spun while worsted yarns are carded, combined, drawn and spun.

In spinning wool fibres, special procedures are required e.g.

Sorting

- The fibres must first be sorted. The fleece is opened and fibres are sorted according to fineness or width and length and sometime according to strength. The grade of fibre determines the type of product for which it will be used.
- Fine fibres that are relatively long are reserved for sheer wool fabrics and worsted medium fibres of short length are suitable for woollen, coarse fibres are made into rough fabrics and carpets.

Scouring

- After sorting the wool is scoured. This involves washing in warm soapy water several times, followed by thorough rinsing and drying.
- Scouring is essential for it removes the natural grease in the fibres, dirt and dust. After scouring the fibres are lubricated for ease in processing.
- For woollen yarns the fibres are fed into the carding machine and carded. They are finally pulled from the card in a form called roving or roping.
- The roving goes directly to spinning and after which it is packed for

Use in weaving and knitting. Woollen yarns are non-uniform bulky, resilience a rough and hairy surface and good insulation properties worsted yarn.

- Following scouring and lubrication, wool fibres are carded.
- The carded sliver is drawn and then combed, after combing the combed sliver is drafted or drawn to form a roving, which is delivered to a spinning frame where the final worsted yarn is made.
- Worsted yarns are relatively smooth and even, highly parallel arrangement, a compact structure, resilient and have good strength.

Newer Methods of Yarn Production

Open –End Spinning

- The term open end spinning has frequently been used as a synonym for all new methods of making yarn. This incorrect, and "open-end" should be applied only to systems where a break occurs in the fibre system.
- Break spinning and open-end spinning are synonymous. The essential steps in open end spinning include the following;
 1. A course sliver of fibres is fed into opening systems which opens the sliver to the point where the fibres are individual entities.
 2. The individual fibres are fed forward.
 3. They are collected together on a small surface and pulled from the surface as a thin layer constantly adding to the forming yarn.
 4. The thin layer is alternated, twist is inserted and the resulting yarn is wound onto bobbins.
 5. Two methods of open-end are in use.
 - (a) Aeromechanical or rotor spinning
 - (b) And fluid or air or water vortex (jet) systems.

Rotor Spinning

- Rotor spinning involves the contact of fibres in or on rotating devices such as funnels, cones, baskets, sieves or needled surfaces.
- A sliver of fibre is fed into the unit, a current of air forces the fibres into a loose form and they are collected on the rotating device.
- As they collect, they are pulled off the rotor by mechanical means to form the yarn and the new yarn is wound onto packages.
- Twist is inserted as the yarn is removed from the rotor.
- Rotor spinning is a popular process and has replaced some ring spinning units.

Air Vortex

- Spinning of yarns by the air vortex system is similar to the rotor process except that the yarn is formed in moving air rather than on a rotating surface.
- Air jet spun yarns are clean, have good uniformity and even quality, are less regular than ring spun yarns, are stronger than rotor spun yarns and have good wash and wear characteristics.

Friction Spinning

- A modification of existing open-end techniques has been introduced that combines mechanical and air current operations. Is generally referred to as friction spinning.
- In this process the sliver enters the system the fibres are separated and spread into a combing or carding roll.
- They are removed from this roll and transported by air to the friction zone.
- Two cylinders rotating of friction spinning drafts through sliver into a center cone of fibre, which is then wrapped by other fibres.
- It is possible to add filament fibres to the centre cone.
- Friction spun yarns are uniform, free of impurities fuller with more body than ring spun yarns but with less strength.

Self-Twist Yarn

- These are essentially two-strand or two-ply (or more) yarns. They may involve one ply or strand of staple fibres and one of filament or two staple fibres or two filament fibres, or multiple combinations of staple and filaments.
- The method is cheap, faster and less labour.
- Yarns are slightly more even and much more stronger than ring spun yarns.

Twist less Yarns

- Fibres are held together with an adhesive to form yarn in the twistless process.
- A roving is made and then drawn to a fine strand. During drawing an adhesive is applied by rollers.
- A common adhesive is starch.
- The drawn yarn, coated with adhesive is wound onto packages, the package is steamed and the adhesive binds the fibres together.
- Twistless yarns stiffer and have greater lusture, less elongation and better covering power than ring spun-yarns.

Core Spun Yarns

- Have a central core with a second layer or sheath of fibres wrapped around it. Both core and wrapping may be of the same fibre type, or two different types.
- Also one may be of filament fibre while the other is of staple fibres or both parts may be either filament or staple-length fibres.
- Core spun yarns are stronger than ring spun yarns.
- They are more even, have good covering powers and have an attractive appearance.

Tape Yarns

- The tape or split film yarns are produced from thin sheets or film of polymer that is cut into strips or ribbons.
- The major steps in production are extraction, cooling, slitting, drawing and winding.
- Tape yarns have good strength, good abrasion resistance and good stability.
- They are used for carpets, sacks and bags, travel goods, some types of ropes etc.

Thread and Yarn

- These are basically similar yarn applied when the assemblage of fibres is employed in the manufacture of fabric.
- Thread indicates a product used to join pieces of fabric together to create a textile product.
- Thread is frequently of plied construction. It is fibre even and strong.

Yarn Classification

Simple and Complex Yarns

- A simple yarn is like in all its parts.
- It can be described as a spun or filament yarn based on the length of fibre present.
- Simple yarns are classified as single ply or cord
- A single yarn is the product of the first twisting operation that is performed by the spinning machine.
- A ply yarn is made by a second twisting operation which combines two or more singles. Each part of the yarn is called ply.
- Plying tends to increase the diameter, strength, uniformity and quality of yarn.

- A cord is made by a third twisting operation, which twists ply yarns together. Some types of sewing thread and some ropes belong to this group. Cord yarns are seldom used in apparel fabrics but are used in industrial weight fabrics such as canvas.

Complex Yarns

- Complex yarns are also referred to as fancy or novelty yarns.
- They may be defined as yarns that have unlike parts and that are irregular at regular intervals.
- The regular intervals may or may not be obvious at first to the observer.
- Fancy yarns may be single, plied or cord yarns.
- They are made for their appearance and value
- Complex yarns are usually composed of base or core an effect and a tie or binder.

Textured Yarns

- Yarns that have been given notably greater apparent volume than conventional yarn of similar type.

Blends Combination Yarns and Fabrics

Blends and combination yarns and fabrics are those in which two or more generic types of fibres are used. The different types can be combined as follows.

1. Two or more different types of fibre can be blended into a single yarn.
 2. Single yarns where each single is of different fibre type can be plied together to form a combination ply yarn.
 3. Single or ply yarns of one fibre type can be used with single or ply yarns of another fibre type and woven or knitted into a fabric
- A blend is an intimate mixture of fabric of different composition, length, diameter, or colour spun together into a yarn.
 - A mixture is a fabric that has yarn of one fibre content in the warp and yarn of different fibre content in the filling.
 - A combination yarn has two unlike strands twisted together as a ply.
 - Blends, mixtures and combinations give properties to fabric that are different from those obtained with one fibre only.

Blending is done for several reasons;

1. To produce fabrics with a better combination of performance characteristics in the product. This is the most important reason for blending.
2. To improve spinning, weaving and finishing efficiency for uniformity of a product.

3. To obtain better texture, hand or fabric appearance.
4. For economic reasons. Expensive fibres can be extended by blending them with more plentiful fibres.
5. To obtain cross-dyed effects or create new colour effects.